Riya Basak

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Profile

My research focuses on world models for general, reliable intelligence: learning structured representations of latent world state, discovering transferable dynamics, reasoning over imagined futures, and acting safely under uncertainty. I study self-supervised representation learning, compositional and causal world modelling, model-based decision-making and safe reinforcement learning, with applications across safety-critical scientific and embodied domains.

Technical Foundations

Programming & Research Computing: Python, C, Java, SQL, Bash, Linux, Git/GitHub, TeX/LaTeX

Mathematical Foundations: Linear Algebra, Probability, Statistics, Calculus, Optimisation

Machine Learning & Representation Learning: Self-Supervised Learning, Representation Learning, Multimodal Learning, CNN–Transformer Architectures, Explainable AI

World Models & Sequential Decision-Making: World Models, Model-Based Prediction, Offline Reinforcement Learning, Safe Reinforcement Learning, Constrained Policy Optimisation, Counterfactual Safety Analysis

Machine-Learning Frameworks: PyTorch, NumPy, scikit-learn, pandas, OpenCV, Matplotlib

Research Methodology: Controlled Baselines and Ablations, Leakage-Aware Validation, Out-of-Distribution Evaluation, Calibration, Predictive-Uncertainty Analysis, Failure-Mode Analysis, Reproducible Experimentation

Education

University of Hertfordshire — **Sep 2023 – May 2026**
BSc (Hons) Computer Science with Artificial Intelligence — First Class Honours — *Hatfield, UK*

GPA: Level 4: **4.34/4.50** | Level 5: **4.34/4.50** | Level 6: **4.44/4.50** | **Cumulative: 4.38/4.50**

Experience

Cubble — **Aug 2025 – Sep 2025**
Full Stack AI Intern — ML Systems Research & Evaluation

- Investigated how deployed AI failures could be inspected systematically rather than through ad-hoc debugging; translated the proposed evaluation methodology into model-review, safety-flag, logging and failure-analysis tools.
- Integrated frontend inspection with backend AI services and structured testing to trace prediction errors, recurring failure modes and application-layer faults reproducibly.
- Improved the research-to-deployment workflow by making model behaviour and failure causes directly inspectable, with clearer experimental documentation and validation.

Cubble — **Jul 2025 – Aug 2025**
Software Engineer Intern — Machine Learning Research

- Investigated which learned visual representations generalised most reliably for safe/unsafe user-uploaded image classification under limited task-specific data; compared ResNet50 and CLIP-based approaches.
- Independently implemented transfer-learning pipelines and evaluated them through controlled validation, optimisation, quantitative error analysis and overfitting analysis to refine model design.
- Achieved **97.5%** accuracy on unseen images with stable validation behaviour; the AI team supervisor recognised my work as the top-performing implementation of the proposed working-model design.

Mercor — **Sep 2025 – Nov 2025**
Artificial Intelligence Researcher — Mathematical Reasoning

- Investigated failure modes in AI mathematical reasoning by formulating Olympiad-level problems as explicit assumptions, constraints, intermediate claims and proof obligations.
- Derived and evaluated solution paths using constructive reasoning, case analysis, edge-case testing and rigorous assumption checking, then assessed AI-generated proofs for validity, completeness and efficiency.
- Converted recurrent errors—including unsupported implications, omitted cases and circular reasoning—into structured evaluation criteria for more precise analysis of mathematical reasoning failures.

Projects

Mitigating Shortcut Learning in Brain Tumour MRI Classification — **Oct 2025 – Mar 2026**
Dissertation Project ***[PREPRINT] [CODE]*** — *Shortcut Learning, Medical Imaging, Explainable AI*

- Proposed Pathology-Focused Disentanglement and Guided Semantic Token Evolution (PFD–GSTE), with variants A and B, for shortcut-learning mitigation without segmentation masks.
- Designed a ResNet50V2–RViT (CNN–Transformer) hybrid for four-class brain-tumour MRI classification; my literature review found no prior peer-reviewed study using this exact combination for the proposed shortcut learning mitigation experiment.
- Trained fixed-split PyTorch models with leakage-aware preprocessing and ablations; all models reached **98.52–99.22%** test accuracy and **98.49–99.20%** macro-F1.
- XAI analysis showed the main result: PFD–GSTE-B gave stronger tumour-region attribution than PFD–GSTE-A, with tumour-focused evidence in about **90%** versus **70%** of reviewed test cases.
- PFD–GSTE-B mitigated shortcut reliance more reliably on the tested benchmark; PFD–GSTE-A was weaker despite high classification accuracy.
- Also performed a small qualitative OOD demonstration on five external T1ce meningioma MRI images, kept separate from the formal benchmark evaluation and project submission.

Admissibility-Aware Offline Actor-Critic Learning for Safer ICU-Sepsis Treatment Decisions — **Mar 2026 – Apr 2026**
Proposed Coursework ***[CODE]*** — *Safe Offline Deep RL*

- Proposed Lagrangian Admissibility-Aware Deep Action-Nudging Actor-Critic (LAADAN-AC), an offline actor-critic method for safer treatment-policy learning in the ICU-Sepsis benchmark.
- Built, to my literature-review findings, a new combination for ICU-Sepsis: hard admissibility masking, twin critics, conservative critic regularisation, expert-policy regularisation, smoothness shaping, and Lagrangian cost control.
- Compared LAADAN-AC with Behaviour Cloning, CQL-regularised fitted-Q, and Vanilla Offline Actor-Critic under the same dataset, seeds, horizon, and evaluation metrics.
- LAADAN-AC achieved the strongest safe policy: **0.7931** ± **0.0007** survival/return, **0.0000** inadmissibility, and the highest expert-action match at **0.9525**.
- LAADAN-AC gave the best return–safety trade-off, improving over BC and CQL while avoiding the unsafe behaviour of unconstrained actor-critic learning.

LAADAN-AC: Beyond Survival in Admissible Offline Treatment-Policy Learning — **Apr 2026 – Jun 2026**
Under review ***[CODE]*** — *Safe Offline Deep RL, Cross-Domain Check*

- Extended LAADAN-AC into a reproducible research manuscript and codebase with component ablations, Lagrangian frontiers, safety-failure analysis, and a cross-domain portability check.

From World Models to Embodied Social Robots — **May 2026 – Present**
Research Project — in Preparation for TMLR — *Embodied AI, Social Robotics, World Models*

- Building a LeWM- and DreamerV3-based world-model stack for embodied agents that learn from predicted physical/social futures, using ARC Prize 2026 / ARC-AGI-3 as an initial abstract testbed before embodied AI and social robotics.
- Propose KCON: Kinetic Counterfactual Oversight Nexus, a compact executive layer between imagined futures and action choice, adding action-effect normlet memory, minimal intervention, uncertainty/novelty handling, and failure-aware selection.

Same Rules, New Worlds — **Jul 2026 – Present**
Research Project (Ongoing) — *World Models, Causal Dynamics, Compositional Generalisation*

- Investigating why world models that predict well in-distribution often fail to transfer learned dynamics across changes in context, appearance, embodiment and long-horizon interaction.
- Developing a mechanism-transport world model that separates persistent state from transferable dynamics, enforcing consistency across nuisance shifts while remaining sensitive to genuine changes in the underlying causal mechanism.
- Evaluating compositional and interventional generalisation under matched appearance–dynamics shifts, with controlled baselines, long-horizon tests and ablations before reporting quantitative claims.